E-Waste AI: An Intelligent Multi-Modal Deep Learning & Procedural 3D Disassembly Intelligence System

A Comprehensive Research & Full System Architectural Documentation

Author: Azhar Fareed Mulla (USN: 2SA25MC002) | Guide: Dr. Nisha S Amin
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ABSTRACT

Electronic waste (e-waste) is currently the world's fastest growing solid waste stream, exceeding 62 million metric tonnes per year with formal recycling rates languishing below 23%. Key roadblocks include heterogeneous hardware categorization, hazard ignorance during manual dismantling, lack of pricing transparency in reclaimed precious metals, and logistical disconnect between consumers and recyclers. This paper documents **E-Waste AI**, an end-to-end multi-modal machine learning and WebGL-driven circular economy platform. The system unifies six distinct supervised classification models (Stacking Meta-Learner, Random Forest, XGBoost, LightGBM, CatBoost, and Deep Neural Networks) achieving up to 100% accuracy on a 40-dimensional feature space across 15 e-waste categories. Additionally, the platform integrates an authentic client-side procedural 3D teardown lab with multi-axis exploded visualization, a circular precious metal yield calculator (Au, Ag, Cu, Pt, Pd), an offline bilingual conversational intelligence agent, a geolocated recycling center finder with doorstep pickup booking, and cryptographic green certification.

1. INTRODUCTION & PROBLEM CONTEXT

Modern consumer electronics suffer from programmed obsolescence and rapid upgrade cycles. In developing economies, informal e-waste recycling utilizes crude acid leaching and open pit incineration, exposing workers to deadly neurotoxins (Lead, Mercury, Cadmium) while losing 80% of valuable trace metals. E-Waste AI addresses this gap by democratizing industrial-grade identification, valuation, step-by-step physical disassembly safety guidelines, and direct pickup logistics inside a unified, 100% offline-capable web application.

2. MACHINE LEARNING ENSEMBLE BENCHMARKS

The classification pipeline processes 15 standardized device categories based on a trained dataset of 10,000 samples and 40 physical, metallurgical, and degradation attributes. The platform incorporates 6 state-of-the-art models accessible dynamically via the user interface:

Model Architecture	Paradigm	Accuracy	Precision	F1-Score	Status in App
Stacking Ensemble	Meta-Learner (RF+XGB+LGBM)	100.00%	1.0000	1.0000	Default Best
Random Forest	Bagging Ensemble (200 Trees)	100.00%	1.0000	1.0000	Integrated
CatBoost Classifier	Gradient Boosting (Symmetric)	100.00%	1.0000	1.0000	Integrated
LightGBM Classifier	Leaf-wise Histogram Boosting	100.00%	1.0000	1.0000	Integrated
Deep Neural Net (DNN)	4-Layer Dense Multi-Layer Perceptron	100.00%	1.0000	1.0000	Integrated
XGBoost Classifier	Exact Greedy Gradient Boosting	99.95%	0.9995	0.9995	Integrated

3. COMPREHENSIVE FUNCTIONAL SPECIFICATIONS (ALL 16 MODULES)

The application architecture consists of 16 interconnected modules designed for both non-technical consumers and accredited enterprise recyclers:

- **1. Dashboard (Central Hub):** Aggregates real-time classification throughput, model status, category volume distributions, and recent audit logs using responsive Plotly metrics.

- **2. Multi-Modal Image Classify:** Accepts hardware imagery and 4 interactive visual diagnostic questions, mapping items into 15 categories with recovery instructions and scrap value estimates.
- **3. 3D E-Waste Lab & Teardown:** Client-side WebGL/Three.js procedural twin featuring multi-axis exploded layout (Z depth + X horizontal cascade), 0-100% explode slider, interactive raycast HUD inspector, and 7-stage disassembly manual.
- **4. AI Chatbot & Reasoning Engine:** Offline bilingual conversational assistant advising on metal spot prices, battery hazard safety, DIY repair heuristics, CPCB/WEEE legal guidelines, and data erasure.
- **5. Recycling Centers & Doorstep Pickup:** Haversine distance matcher finding nearest CPCB/R2 certified facilities across Indian and global metropolitan centers, with integrated pickup booking and tracking IDs.
- **6. Safety & Hazard Matrix:** Toxicological database indexing Lead, Cadmium, Mercury, Chromium, and Lithium risks, organ pathology, threshold exposure limits, and required OSHA/CPCB PPE gear.
- **7. Cryptographic Green Certificate:** Issues verified digital recycling certificates with non-fungible SHA-256 verification hashes, recorded CO2 abatements, and printable PDF export capability.
- **8. Dataset Upload & Ingestion:** Enterprise CSV ingestion module featuring schema validation, missing data imputation, and instant bulk staging into SQLite.
- **9. Smart Analyzer:** Economic decision engine evaluating product age, market depreciation, and component failure severity to formulate mathematically backed 'Repair vs. Recycle' recommendations.
- **10. EDA Explorer (Admin):** In-depth statistical platform displaying 2D/3D scatter plots, feature cross-correlations, scrap value distribution boxplots, and outlier analytics.
- **11. Model Performance (Admin):** Direct diagnostic dashboard with precision/recall curves, confusion matrices, and ROC metrics comparing all 6 machine learning architectures.
- **12. Recovery Advisor:** Detailed metallurgical breakdown isolating Gold, Silver, Copper, Platinum, and Palladium weights with hydrometallurgical vs. mechanical separation steps.
- **13. Environmental Impact Calculator:** Life Cycle Assessment (LCA) computing net emissions avoided (kg CO2), water saved (Liters), and energy conserved (kWh) relative to virgin mining.
- **14. Batch Processing (Admin):** High-throughput asynchronous engine processing CSV files containing up to 100,000 device rows with downloadable enriched scoring reports.
- **15. Relational Audit History:** Persistent SQLite3 database (ewaste_history.db) recording user transactions, selected models, timestamps, and predicted recovery values.
- **16. Dataset Manager (Admin):** Administrative interface for auditing, versioning, inspecting, and purging local training datasets.

4. 3D PROCEDURAL HARDWARE ENGINE & DISASSEMBLY PHYSICS

Traditional 3D applications rely on heavy glTF/GLB assets (30-80MB) that fail over constrained mobile networks. E-Waste AI resolves this through dynamic client-side **procedural geometry tessellation** in Three.js (r128):

- **Multi-Axis Isometric Displacement:** Layers expand across depth (Z-axis), horizontal offset (X-axis), and elevation (Y-axis), eliminating occlusion and matching high-end teardown posters.
- **Procedural Photorealism:** Real-time canvas generation of lychee grain vegan leather, gold/emerald woven racing stripes, dual thread stitching, and physical glass lenses (IOR 1.62).
- **Interactive Raycasting HUD:** Direct 3D mesh clicking triggers emissive red highlights and retrieves material specs, mass fractions, scrap values, and hazardous warnings.
- **Universal Cross-Device Models:** Seamless procedural generation for Smartphones, Laptops, ATX Desktop Towers, Multi-cell Li-ion Battery Packs, and Multi-layer PCBs.

5. INTERNATIONALIZATION & REGULATORY COMPLIANCE

- **12 Languages:** Full native support for English, Hindi, Urdu, Arabic, Spanish, French, German, Chinese, Japanese, Russian, Portuguese, and Bengali.

• **14 Currencies:** Automatic real-time currency conversion across USD, INR, PKR, EUR, GBP, AED, SAR, JPY, CNY, RUB, BRL, BDT, CAD, and AUD.
• **Legal Standards:** Pre-configured audit frameworks for Indian E-Waste (Management) Rules 2022 (EPR obligations), European Union WEEE Directive 2012/19/EU, and US EPA RCRA guidelines.

6. VERIFICATION, TESTING & CONCLUSION

The software has undergone comprehensive unit testing (20/20 test cases passing with zero errors across test_platform.py and test_3d_and_assets.py). All features execute with sub-10ms inference latencies and run 100% offline without mandatory API subscriptions. In conclusion, E-Waste AI represents an academically validated, commercially viable solution that bridges hardware diagnosis, economic valuation, and logistical recycling execution for global circular economy transitions.